

PwC Inclusive Future Lab – Project Final Report

MediPort — A Patient-Owned Disease Timeline and Consultation Communication Tool for People with NF2

Team: CareLink Lab

Project period: 20 July – 30 August 2026

Members: Wanting (Echo) Zhao, *Project Lead* · Yiran Sun · Shengxuan Zhang

Part I — Project Information

1 Background

1.1 Context and rationale

Neurofibromatosis Type 2 (NF2) is a rare autosomal-dominant disorder characterised by bilateral vestibular schwannomas and other tumours of the brain and spine. Care is lifelong and highly multidisciplinary. Because clinical decisions depend heavily on changes across serial MRI scans and consultations, a patient's longitudinal history is often more important than any single medical record.

In China, the diagnosis and treatment continuity of NF2 meets with challenges. There is no dedicated national NF2 specialist centre, so patients may receive care across institutions and provinces. Hospital information systems remain largely institution-specific, leaving patients responsible for carrying paper reports, MRI films, and their own recollection from one consultation to the next.

Communication adds another layer of difficulty. Progressive sensorineural hearing loss is a defining feature of NF2. In a busy consultation, background noise, masks, time pressure, and reliance on spoken communication can make it difficult for patients to communicate directly with clinicians and to receive information accurately.

1.2 Problem definition

Our research reframed the challenge from **record storage to information transmission**. The information clinicians need often already exists, but it is trapped in paper records, imaging films, fragmented hospital systems, or unaided patient recall. The patient must then reconstruct years of medical history during a short consultation.

Problem statement. People with NF2 in China must compress a longitudinal, multi-institutional medical history into a short consultation, sometimes through a communication channel that the disease itself is progressively impairing. The information exists, but there is no patient-owned artefact that can reliably carry it into the clinical conversation.

Design question. How might we help patients own, carry, and communicate their longitudinal medical history across institutions within seconds, without relying on hearing, vision, or mobility?

This framing established the consultation—not the archive—as the primary design context. Accessibility therefore became a core product requirement rather than an adaptation added later.

1.3 Research findings

We conducted a survey in a Chinese NF2 patient WeChat community from 28 June to 6 July, receiving 38 valid responses, followed by four semi-structured asynchronous interviews with two patients and two caregivers. Detailed findings are provided in Appendices 2 and 3.

1.3.1 Limited use of digital health tools.

None of the 38 respondents used a structured digital tool to manage their medical history. 65.8% relied on paper records and MRI films, while 34.2% had no systematic method. Since 89.5% had been diagnosed for three or more years, these appear to be established practices rather than temporary workarounds.

1.3.2 Fragmented information across care settings.

71.1% reported difficulty when seeing a new doctor or seeking care across hospitals. The main barriers were difficulty explaining a fragmented history within a short consultation (36.8%) and insufficient consultation time for clinicians to understand years of disease progression (28.9%).

1.3.3 Accessibility barriers

86.8% reported at least one impairment that could affect interaction with digital or clinical systems: hearing loss (76.3%), motor difficulty (39.5%), balance difficulty (39.5%), and vision impairment (28.9%).

1.3.4 Information asymmetry during consultations

Across all four interviews, a hearing family member had acted as an interpreter or communication bridge. One participant reported that clinicians appeared reluctant to address him directly. This finding shifted our understanding of the summary: it was

not simply a time-saving tool, but potentially an accessibility mechanism that could restore a more direct patient–clinician interaction.

1.4 Scope revision

As submitted	As executed	Rationale
AI medical-record ingestion as a core innovation	Roadmap capability; not implemented in the prototype	Reliable extraction could not be validated against real medical records within six weeks.
Community, research digest, and health-record modules	Static concept cards and roadmap items	Scope was narrowed to deepen the core consultation pathway.
Five core screens	31 high-fidelity frames	A realistic end-to-end experience required onboarding, accessibility setup, record capture, and the three-tab consultation summary.

2 Implementation Process

2.1 Collaboration

The team used flexible weekly check-ins, daily asynchronous communication, and shared working documents through Feishu. Work was coordinated through an execution tracker and reviewed against actual delivery at project close.

2.2 Six-week execution

Week 1 (20–26 Jul) — Define

Reviewed research findings and empathy assets; converged four personas into one primary and two secondary personas; created the problem-statement canvas; established the Figma structure and initial design tokens.

Week 2 (27 Jul–2 Aug) — Ideate

Ran a Crazy Eights workshop and generated approximately 30 solution ideas; clustered and prioritised concepts; completed competitor review, information architecture, user flows, low-fidelity wireframes, and accessibility research.

Week 3 (3–9 Aug) — Build

Submitted the interim deliverable; established the high-fidelity design system; refined information architecture and user flows; began the consultation summary.

Week 4 (10–16 Aug) — Build

Completed the medical summary concept; developed the high-fidelity prototype; refined individual screens; began the clickable prototype; prepared an accessible usability-testing script.

Week 5 (17–23 Aug) — Validate

Completed the clickable prototype; reviewed the design against an accessibility checklist; resolved identified issues; prepared the final design package.

Week 6 (24–30 Aug) — Finalise

Refined the final report and prepared the presentation deck.

3 Outcomes

3.1 High-fidelity interactive prototype — 31 frames

The prototype covers the core journey from first launch to consultation preparation. See Appendix 1 for the complete prototype.

Entry and onboarding

The experience begins with language selection (Simplified Chinese / English), a concise value proposition, registration, and verification.

Accessibility-first onboarding

Before general registration, users configure an accessibility profile across four dimensions:

- **Vision:** larger text and higher contrast

- **Hearing:** captions, visual alerts, and vibration
- **Touch:** larger targets and tolerance for motor difficulty
- **Simplicity:** fewer steps and clearer guidance for users who prefer a lower-complexity experience

Users then select a patient or caregiver role.

The prototype also introduces an **AID Card** entry point (AID Card function developing) designed for emergency situations so that relevant accessibility and medical information can be surfaced quickly on the user's device.

This sequence deliberately places accessibility before general account setup: users should not have to navigate an inaccessible experience before being given the tools to adapt it.

Home

The primary action is **Upload**, allowing users to capture medical records with the camera or import files from the device.

After upload, users can save the record or generate an AI-assisted summary.

Secondary actions include:

- **Log Symptoms**
- **My Notebook**
- **AI Summary**

Health-resource cards provide contextual information without competing with the core workflow.

Consultation Summary — the central experience

The consultation summary is organised into three tabs:

- **Snapshot**
- **Symptoms**
- **Medical Background**

The Snapshot prioritises information a clinician needs quickly:

- 1) Patient identity

- 2) Communication accommodations, such as facing the patient when speaking and confirming important information in writing
- 3) Current concerns, based on a short pre-consultation self-check
- 4) Key clinical facts
- 5) A longitudinal medical timeline, with source reports and imaging attached to individual events
- 6) NF2 related symptoms screening
- 7) Other clinician-interested medical information, including past medical, surgical, medication, allergy, and family history

The design turns a fragmented record collection into a clinician-readable narrative while preserving links to the underlying evidence.

Concept cards and roadmap

Additional concepts include:

- A privacy-conscious patient community
- A **Care Hub** for disease information and support resources
- A year-based document archive

These remain concept-level features rather than core prototype flows.

3.2 Supporting design assets

The project also produced the supporting UX artefacts required to explain and validate the product direction:

- Research reports
- Personas
- Journey maps
- Empathy maps
- Problem statements
- POV boards
- Ideation outputs

- **Information architecture**
- **User flows**
- Design-system components
- Competitors Audit
- Accessibility criteria¹

See Appendices 2–12.

4 Innovation

The product direction addresses three connected problems: **fragmented artefacts, inaccessible communication channels, and the need for trustworthy information at the point of care.**

4.1 Accessibility by design

Accessibility is introduced at first launch and carried through the design system rather than treated as a final compliance check.

Visual, auditory, motor, and cognitive-accessibility considerations inform component behaviour, interaction patterns, and information presentation.

4.2 Designing for an underserved population

The project focuses on a rare-disease population that is poorly served by general-purpose health-management systems.

Rather than attempting to replace hospital information systems, MediPort gives patients ownership of the continuity layer between institutions.

¹This project has followed the guidance of **Web Content Accessibility Guidelines (WCAG) 2.2**.
<https://www.w3.org/TR/2024/REC-WCAG22-20241212/>

4.3 A communication channel designed for hearing-impaired patients

For many people with NF2, hearing loss changes the conditions of a medical consultation.

The consultation summary therefore acts as a communication interface: it presents essential information in a clear, structured, text-based format that both patient and clinician can reference.

The summary structure is based on the information clinicians need to understand a patient's current situation and longitudinal progression, rather than simply reproducing the structure of hospital records.

This creates an opportunity to reduce communication friction while supporting faster clinical orientation and decision-making.

5 Social Impact

Individual

The immediate potential impact is to strengthen the patient's position in the consultation. Where a family member may currently mediate communication, a patient-owned summary provides a shared visual and textual reference that can support more direct interaction between patient and clinician.

Community

The underlying design pattern is not limited to NF2. Longitudinal, cross-institutional care is common in oncology, diabetes, multimorbidity and other chronic diseases. Communication barriers also affect people with hearing, speech, cognitive, or other accessibility needs.

The disease-specific content can change while the underlying information-transmission model remains applicable.

6 Limitations

The project remains at the concept-validation stage.

The prototype has not been released into clinical use, and no real-world usage or outcome data are available. Formal usability testing and clinical validation remain future steps.

Accordingly, our impact claims are limited to two areas:

- 1) Documenting and quantifying the problem through the completed research.
- 2) Demonstrating a coherent, accessible, and clinically credible design response through the prototype.

7 Future Outlook

The project suggests a broader opportunity for patient-owned health management in rare and chronic diseases.

Potential extensions include:

- Remote-consultation preparation
- Plain-language explanations of treatment options
- Clinical-trial alerts
- Home-care support
- Cross-institutional care coordination

The next stage of research will investigate whether the communication and coordination challenges identified in NF2 also appear across other rare and chronic conditions, and which elements of the MediPort model can be generalised into a broader patient-owned health-management solution.

Part II — Team Roles and Member Contributions

Member	Role
Zhao Wanting (Echo)	Project lead; product and experience design; user research
Sun Yiran	Product design; accessibility review; documentation
Zhang Shengxuan	Medical content; user research

1 Zhao Wanting (Echo) — Project Lead

Responsibilities

Project planning, scheduling, and risk management; survey design, distribution, and analysis; four interviews and associated research reports; facilitation of the point-of-view workshop, persona reduction, problem statements, solution categorisation, and goal definition; co-development of information architecture, user flows, the design system, low-fidelity wireframes, and the high-fidelity clickable prototype; final reporting.

Capability development

From data science to design decisions.

I leveraged my data science background to bring a quantitative perspective to UX research. I designed and analysed the survey and interview, translated patterns in the data into actionable user insights, and used evidence to prioritise design decisions. This experience strengthened my ability to connect data analysis with human-centred design rather than treating quantitative research as a separate stage.

UX design practice

I applied methods from the Google UX Design Certificate to a real-world rare-disease problem, moving from research and empathy work through problem definition, ideation, information architecture, interaction design, and validation.

The process strengthened my ability to consider the user's experience across the complete journey rather than treating individual screens as isolated deliverables.

Figma and prototyping

I developed the low- and high-fidelity designs, design-system components, and clickable prototype through iterative self-learning and consultation with lecturers when technical or interaction-design problems arose.

Project management and risk mitigation

I coordinated the project scope, milestones, dependencies, and deliverables.

When the six-week timeline made some proposed features impractical to validate, I prioritised the consultation pathway and moved lower-confidence capabilities, such as automated medical-record ingestion, into the roadmap rather than presenting them as validated product functionality.

2 Sun Yiran — Product Design, Accessibility Review, Documentation

Specific division of work and individual contributions

1. Information accessibility research and evaluation:

I collaborated with team members to lead the research on information accessibility, the formulation of reference standards, conceptual design, and final evaluation.

2. Preliminary research and competitive analysis for conceptual design:

Preliminary research: I participated in discussions on needs insights and the construction of user personas;

Conceptual design: I was involved in the review of information architecture and low-fidelity design.

3. Meeting minutes, data organization, presentation material production, and report writing and formatting:

Meeting minutes: I recorded meeting minutes during weekly meetings, ensuring traceability of communication records;

Data organization: I organized process documents and final deliverables before producing presentation materials;

Presentation material production: I collaborated with team members to lead the production of mid-term presentation materials and the Chinese version of the final report.

Personal growth

1. I shifted my focus from physical accessibility to information accessibility, made friends with teammates from different majors, identified my positioning and strengths, and built confidence;

2. I learned about the principles of interaction design, enabling me to better use software and information technology to manage my own life.

3 Zhang Shengxuan — Medical Content and User Research

Individual Contribution and Skills Development

Preliminary Research

I participated in questionnaire design and data collection, and took the lead in conducting user interviews to gain a deeper understanding of the experiences, difficulties, and needs of the target users. I also led the investigation of publicly available medical information related to the project background, providing reliable references for subsequent needs analysis and design decisions.

Through this process, I improved my ability to conduct user research, organize qualitative information, identify meaningful insights from user feedback, and evaluate the reliability and relevance of supporting information.

Concept Development

I participated in needs analysis and the development of user personas, and took the lead in designing the information architecture and user flows. I was responsible

for organizing fragmented user needs into a coherent product structure and mapping the key steps users would take when accessing and completing core tasks.

This process strengthened my ability to translate research findings into concrete design requirements and helped me develop a more systematic understanding of product structure and user journeys.

Design Practice

I led the design of the **Visit Summary** feature, including the organization and writing of example content. I also participated in page layout planning and the development of accessibility-related features. Through this work, I further developed my ability to structure and visualize complex medical information in a way that is easier for users to understand and use.

I also gained a stronger awareness of designing interfaces according to users' different abilities and practical usage barriers.

Overall Process

Throughout the project, I was also responsible for reviewing the accuracy of medical information and reducing potential misunderstandings caused by inaccurate or inappropriate expressions.

Overall, this project helped me develop a more complete UX design process, from user research and needs analysis to information architecture, user flow, and high-fidelity interface design. It also strengthened my ability to apply user-centered thinking and professional information judgment in the design of healthcare-related digital products.

Appendices

No.	Title	Format
Appendix 1	High-fidelity interactive prototype, 31 frames	PDF + PNG + MOV ²
Appendix 2	Survey findings report (n = 38), interview analysis, charts	PDF + XLSX + MD + PNG
Appendix 3	Interview findings report (n = 4), transcripts	PDF
Appendix 4	Personas, journey maps, empathy map, user stories	PNG
Appendix 5	Problem statements, POV board	PNG
Appendix 6	Ideation, categorisation, sketches	PNG
Appendix 7	Clinical Communication Brief	PDF
Appendix 8	Six-week plan and execution tracker	MD + XLSX
Appendix 9	Information architecture	PNG
Appendix 10	User flow	PNG
Appendix 11	Accessibility Competitors Audit	PDF

² Video could also be viewed on [Gohttps://drive.google.com/file/d/1VeSNdxPfDrgpclxiK1_76f4eyNPoFTJ1/viewogle](https://drive.google.com/file/d/1VeSNdxPfDrgpclxiK1_76f4eyNPoFTJ1/viewogle)

No.	Title	Format
Appendix 12	Accessibility Review	MD + XLSX

Prepared by CareLink Lab. All research findings and design outputs are traceable to the primary materials listed in the appendices. Items that remain incomplete or conceptual are explicitly identified.